

Out($\mathrm{GL}_2(\mathbb{F}_p)$)

Aidan Hennessey

April 2025

1 Motivation and investigation

This summer, I participated in a research project where we computed a certain invariant attached to smooth curves of the form $y^2 = f(x)$ for f of degree 5 or 6. The invariant we computed took the form of a matrix subgroup of $\mathrm{GL}_4(\mathbb{F}_5)$. In order to interpret our results, we needed to develop nice descriptions of various matrix groups from opaque generating sets.

Some of the more difficult to understand matrix groups were a family of matrix groups each consisting of all block-upper-triangular matrices such that the two diagonal blocks differ by a fixed automorphism. We only cared about the matrix groups up to conjugation, so it only mattered to us what the automorphism was up to conjugation. The discovery of these matrices in our results motivated the following natural question: what are the outer automorphisms of $\mathrm{GL}_2(\mathbb{F}_p)$?

A guess at the structure of this group may be informed by the following classical result.

Theorem 1 (The continuous setting). In the category of lie groups, $\mathrm{Out}(\mathrm{GL}_2(\mathbb{C})) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, generated by $M \mapsto (M^{-1})^\top$ and $M \mapsto \overline{M}$, where $\overline{M}$ is the matrix obtained by replacing each entry of M with its complex conjugate.

Based on this, an unfamiliar observer may similarly expect $\mathrm{Out}(\mathrm{GL}_2(\mathbb{F}_p)) \cong \mathbb{Z}/2\mathbb{Z}$, generated by inverse transpose. Here is the output of the corresponding Magma code for $p = 5$:

```
≡ In [53]: 1 G := GL(2, 5);
           2
           3 // #Out = #Aut / # Inn
           4 #AutomorphismGroup(G) / (#(G/Center(G)));

Out[53]: 4
```

Figure 1: A computation of the order of $\mathrm{Out}(\mathrm{GL}_2(\mathbb{F}_5))$

Huh, weird. These kinds of things shouldn't be too surprising though; small linear groups are known for sometimes having exceptional properties that go away for large p . Here is the data for (the hopefully better behaved) larger primes.

```

≡ In [55]: 1 for p in PrimesInInterval(5, 100) do
           2   G := GL(2, p);
           3   // #Out = #Aut / # Inn
           4   out_size := #AutomorphismGroup(G) / (#G/#Center(G));
           5   printf "GL(2, %o) has outer auto group of order %o\n", p, out_size;
           6 end for;

Out[55]: GL(2, 5) has outer auto group of order 4
         GL(2, 7) has outer auto group of order 4
         GL(2, 11) has outer auto group of order 8
         GL(2, 13) has outer auto group of order 8
         GL(2, 17) has outer auto group of order 16
         GL(2, 19) has outer auto group of order 12
         GL(2, 23) has outer auto group of order 20
         GL(2, 29) has outer auto group of order 24
         GL(2, 31) has outer auto group of order 16
         GL(2, 37) has outer auto group of order 24
         GL(2, 41) has outer auto group of order 32
         GL(2, 43) has outer auto group of order 24
         GL(2, 47) has outer auto group of order 44
         GL(2, 53) has outer auto group of order 48
         GL(2, 59) has outer auto group of order 56
         GL(2, 61) has outer auto group of order 32
         GL(2, 67) has outer auto group of order 40
         GL(2, 71) has outer auto group of order 48
         GL(2, 73) has outer auto group of order 48
         GL(2, 79) has outer auto group of order 48
         GL(2, 83) has outer auto group of order 80
         GL(2, 89) has outer auto group of order 80
         GL(2, 97) has outer auto group of order 64

```

Figure 2: Uh oh...

Rather than settling down to 2 as p grows, the size of the outer automorphism group keeps increasing roughly linearly in p . Clearly, these automorphisms are not some sporadic, exceptional coincidences. Indeed, we will now explain where they are coming from.

2 Radial automorphisms

Theorem 2 (Radial automorphisms). The map $\rho_n : M \mapsto (\det M)^n M$ is an automorphism of $\mathrm{GL}_2(\mathbb{F}_p)$ exactly when $\gcd(2n+1, p) = 1$. Such automorphisms are called *radial automorphisms*. Distinct radial automorphisms represent different elements of the outer automorphism group.

Proof. The map $\rho_n : M \mapsto (\det M)^n M$ is readily seen to be a homomorphism for any general linear group, by virtue of the determinant being multiplicative and scalar matrices being central. We thus just have to determine when ρ_n is a bijection (equivalently, a surjection, since $\mathrm{GL}_2(\mathbb{F}_p)$ is finite). Notice that ρ_n restricts to the identity on $\mathrm{SL}_2(\mathbb{F}_p)$, so the map ρ_n is surjective if and only if its composition with the quotient $\mathrm{GL}_2(\mathbb{F}_p) \twoheadrightarrow \mathrm{GL}_2(\mathbb{F}_p)/\mathrm{SL}_2(\mathbb{F}_p) \cong \mathbb{F}_p^\times$ is a surjection. The composition $\det \circ \rho_n$ sends $M \mapsto (\det M)^{2n+1}$. Thus, ρ_n is a surjection if and only if the $(2n+1)$ th power map on $\mathbb{F}_p^\times$ is a surjection. The group $\mathbb{F}_p^\times$ is cyclic of order $p-1$, so the $(2n+1)$ th power map is surjective if and only if $\gcd(2n+1, p-1) = 1$, as desired.

Conjugation preserves determinant. Thus, since each ρ_n yields a different composition with the determinant map, each distinct ρ_n represents a distinct element of $\text{Out}(\text{GL}_2(\mathbb{F}_p))$. To see the compositions with the determinant are distinct, let α be a generator of $\mathbb{F}_p^\times$, and consider $\begin{bmatrix} 1 & 0 \\ 0 & \alpha \end{bmatrix}$. $\square$

By Fermat's little theorem, two values n, m give the same radial automorphism exactly when $n \equiv m \pmod{p-1}$, equivalently, when $2n+1 \equiv 2m+1 \pmod{2(p-1)}$. All units mod $2(p-1)$ are odd, so the above result gives a family of $\phi(2(p-1))$ outer automorphism classes, where ϕ is the Euler totient function. Let's see how this compares to the previously computed data.

```

In [58]: 1 for p in PrimesInterval(5, 100) do
          2 EulerPhi(2*(p-1));
          3 end for;

Out[58]: 4
          4
          8
          8
          16
          12
          20
          24
          16
          24
          32
          24
          44
          48
          56
          32
          40
          48
          48
          48
          80
          80
          64

```

Figure 3: The numbers of outer automorphism classes of $\text{GL}_2(\mathbb{F}_p)$ represented by radial automorphisms, for increasing p .

The astute reader will observe that this output exactly matches the originally computed size of $\text{Out}(\text{GL}_2(\mathbb{F}_p))$. That motivates one to conjecture the following theorem, which we will now prove.

Theorem 3 (It's all radial). $\text{Out}(\text{GL}_2(\mathbb{F}_p)) \cong (\mathbb{Z}/(2(p-1))\mathbb{Z})^\times$, with isomorphism given by $(M \mapsto (\det M)^n M) \leftrightarrow n$.

Proof. Injectivity has already been established. It remains to show surjectivity, i.e. that every automorphism of $\text{GL}_2(\mathbb{F}_p)$ is radial, up to conjugation.

We first fix a(n admittedly large) body of notation. For M a matrix, let f_M denote its characteristic polynomial. Let U^+ denote the subgroup of $\text{GL}_2(\mathbb{F}_p)$ consisting of upper-triangular matrices with 1's on the diagonal. Similarly, let U^- denote the subgroup consisting of lower triangular matrices with 1's on the diagonal. Let B^+ and B^- denote the groups of upper- and lower-triangular matrices, respectively. As the notation may suggest, U^+ and U^- are conjugate subgroups. We refer to subgroups conjugate to U^+ as *unipotent*. Similarly, we refer to the class of subgroups conjugate to

B^+ as the *Borel subgroups*. The Borel subgroups are very special. They are the stabilizers of 1-D subspaces of $\mathbb{F}_p^2$, they are the normalizers of unipotent groups, and they are maximal subgroups of GL_2 . Finally, let $H = B^+ \cap B^-$ denote the group of diagonal matrices. The group H and its conjugates are known as *split Cartan* subgroups of GL_2 . Split Cartan subgroups are index 2 in their normalizers; $N(H)$ consists of all matrices which are diagonal or anti-diagonal.

We now state and prove this result's workhorse lemma.

Lemma 1 (Automorphisms of $\mathrm{SL}_2(\mathbb{F}_p)$). Let σ be an automorphism of $\mathrm{GL}_2(\mathbb{F}_p)$. Then, σ acts on SL_2 by conjugation by an element of GL_2 .

Proof. Let $\sigma \in \mathrm{Aut}(\mathrm{GL}_2(\mathbb{F}_p))$. The unipotent groups are Sylow- p subgroups, so they are permuted by σ . Similarly, the Borel subgroups, by virtue of being the normalizers of the Sylow p -subgroups, must be permuted by σ . Let $\sigma(B^+) = B_1$ and $\sigma(B^-) = B_2$. The groups B_1 and B_2 are characterized as stabilizers of 1-D subspaces ℓ_1 and ℓ_2 . Let P be a matrix taking the standard basis to generators of ℓ_1 and ℓ_2 . Conjugating B_1 and B_2 and P yields B^+ and B^- , respectively. Thus, by composing with the appropriate conjugation, we may normalize σ to fix B^+ and B^- .

Each Borel group contains only a single Sylow p -subgroup, so we conclude that U^+ and U^- are fixed by σ . Similarly, we see that $H = B^+ \cap B^-$ is fixed, and thus $N(H)$, and even $N(H) \setminus H$, are fixed. From this, we have the following forms for the images of three special matrices.

$$\sigma \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \quad \sigma \left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \quad \sigma \left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & c \\ c^{-1} & 0 \end{bmatrix}$$

The restrictions on the images of $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Y = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ come from the fact that σ fixes U^+ and U^- . The restrictions on the image of $Z = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ come from the fact that σ preserves $N(H) \setminus H$, and that $\sigma(Z)$ must have order 2. These matrices are important because X and Y generate $\mathrm{SL}_2(\mathbb{F}_p)$ and conjugation by Z swaps X and Y . Since $ZXZ = Y$, we must have $\sigma(Z)\sigma(X)\sigma(Z) = \sigma(Y)$. Computing, this gives

$$\begin{bmatrix} 0 & c \\ c^{-1} & 0 \end{bmatrix} \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & c \\ c^{-1} & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ c^{-2}a & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}.$$

Thus, we conclude $a = bc^2$. Now observe that H normalizes U^+ , U^- , B^+ , B^- , and itself, so we are free to further normalize σ by conjugating by an element of H . Conjugating by $\begin{bmatrix} b & 0 \\ 0 & 1 \end{bmatrix}$ has the effect of setting $b = 1$. To show σ fixes $\mathrm{SL}_2(\mathbb{F}_p)$, it thus remains to show that $c^2 = 1$. For $p = 2, 3$, this is immediate, as all units mod p square to 1. For $p \geq 5$, we will use the following proposition.

Proposition 1. Let $p \geq 5$. A matrix $M \in \mathrm{SL}_2(\mathbb{F}_p)$ has order 6 if and only if its characteristic polynomial is the 6th cyclotomic polynomial $\Phi_6(\lambda) = \lambda^2 - \lambda + 1$.

Proof. In characteristics other than 2 and 3, Φ_6 has distinct roots, and thus any matrix with characteristic polynomial Φ_6 has minimal polynomial Φ_6 . By construction, Φ_6 first divides $x^n - 1$ when $n = 6$, so if $f_M = \Phi_6$, then M has order 6.

Conversely, suppose M is order 6. If M were not diagonalizable, then M would have order a multiple of p , so (because $p \geq 5$), M is diagonalizable over $\overline{\mathbb{F}}_p$. The eigenvalues of M are 6th roots of unity. Because $M \in \mathrm{SL}_2$, the eigenvalues are reciprocal roots of unity. In order for M not to have

order properly dividing 6, these roots of unity must be primitive. We thus conclude $M \simeq \begin{bmatrix} \zeta_6 & 0 \\ 0 & \zeta_6^{-1} \end{bmatrix}$, and thus that $f_M(\lambda) = \lambda^2 - \lambda + 1$. $\square$

Corollary 1. If σ is an automorphism of $\mathrm{GL}_2(\mathbb{F}_p)$ for $p \geq 5$, then σ preserves the set of matrices with characteristic polynomial Φ_6 .

With corollary in hand, compute that

$$\sigma(X)\sigma(Y)^{-1} = \begin{bmatrix} 1 & c^2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1-c^2 & c^2 \\ -1 & 1 \end{bmatrix}.$$

Setting $c^2 = 1$ recovers XY^{-1} . Taking the characteristic polynomial, we see that $f_{XY^{-1}}(\lambda) = \lambda^2 - \lambda + 1$. By the corollary, $f_{\sigma(X)\sigma(Y)^{-1}}(\lambda) = \lambda^2 - \lambda + 1$ for $p \geq 5$. On the other hand, we may compute directly that $f_{\sigma(X)\sigma(Y)^{-1}}(\lambda) = \lambda^2 - (2-c^2)\lambda + 1$. Equating coefficients yields that $c^2 = 1$, as desired.

In summary, we have shown that for $\sigma \in \mathrm{Aut}(\mathrm{GL}_2(\mathbb{F}_p))$ an arbitrary automorphism, there is a representative of the coset $[\sigma] \in \mathrm{Out}(\mathrm{GL}_2(\mathbb{F}_p))$ which acts as the identity on $\mathrm{SL}_2(\mathbb{F}_p)$. This completes the proof of the main lemma. $\square$

It remains to show that an automorphism of $\mathrm{GL}_2(\mathbb{F}_p)$ which acts as the identity on $\mathrm{SL}_2(\mathbb{F}_p)$ must be radial. Indeed, let σ be such an automorphism. Let α be a generator of $\mathbb{F}_p^\times$. Letting $K = \langle \begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix} \rangle$, one may check that $\mathrm{GL}_2(\mathbb{F}_p) = K \rtimes \mathrm{SL}_2(\mathbb{F}_p)$. In other words, every matrix $M \in \mathrm{GL}_2(\mathbb{F}_p)$ admits a unique decomposition $M_{\det} M_s$ where $M_{\det} = \begin{bmatrix} \det M & 0 \\ 0 & 1 \end{bmatrix}$ and $M_s = M_{\det}^{-1} M$ has determinant 1. Thus, σ is determined by what it does to $\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}$.

The unipotent groups lie in $\mathrm{SL}_2(\mathbb{F}_p)$, and thus are fixed by σ . Thus, σ restricts to an automorphism of each Borel subgroup, and thus of each split Cartan subgroup. Therefore, $\sigma(\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}) \in H$. Let $\sigma(\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}) = \begin{bmatrix} \alpha^n & 0 \\ 0 & \alpha^m \end{bmatrix}$. Towards relating n and m , observe that the conclusion above that $c^2 = 1$ implies that the image of $Z = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ under σ is either Z or $-Z$. In particular, conjugation by Z commutes with σ . Recall that $\begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} \in \mathrm{SL}_2(\mathbb{F}_p)$, so $\sigma(\begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix}) = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix}$. Now, compute that

$$\begin{aligned} \begin{bmatrix} \alpha^n & 0 \\ 0 & \alpha^m \end{bmatrix} &= \sigma \left(\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix} \right) \\ &= \sigma \left(\begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \alpha \end{bmatrix} \right) \\ &= \sigma \left(\begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} Z \begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix} Z \right) \\ &= \begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} \cdot Z \cdot \sigma \left(\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix} \right) \cdot Z \\ &= \begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} \cdot Z \cdot \begin{bmatrix} \alpha^n & 0 \\ 0 & \alpha^m \end{bmatrix} \cdot Z \\ &= \begin{bmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{bmatrix} \begin{bmatrix} \alpha^m & 0 \\ 0 & \alpha^n \end{bmatrix} \\ &= \begin{bmatrix} \alpha^{m+1} & 0 \\ 0 & \alpha^{n-1} \end{bmatrix}. \end{aligned}$$

We conclude that $n = m+1$, so $\sigma\left(\begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}\right) = \alpha^m \begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}$. We see that σ agrees with a radial isomorphism, namely $M \mapsto (\det M)^m M$, on a generating set for $\mathrm{GL}_2(\mathbb{F}_p)$, and thus σ is radial. $\square$

3 The general picture

The structure of the outer automorphism group of $\mathrm{GL}_2(\mathbb{F}_p)$ is a special case of a more general result about general linear groups over fields. To state the result properly, we must first generalize the definition of radial automorphism.

Definition 1 (Radial automorphism of a group). A radial automorphism of a group G is one of the form $g \mapsto \chi(g)g$, where $\chi : G \rightarrow Z(G)$ is some character (homomorphism) from G to its center.

The case of $\mathrm{GL}_2(\mathbb{F}_p)$ is abnormally simple in that $\mathbb{F}_p$ lacks field automorphisms. Given an automorphism $\sigma : K \rightarrow K$, one obtains an automorphism $\tilde{\sigma} : \mathrm{GL}_n(K) \rightarrow \mathrm{GL}_n(K)$ via applying σ to each entry of each matrix. Nontrivial automorphisms σ necessarily induce automorphisms $\tilde{\sigma}$ which act nontrivially on the center $Z(\mathrm{GL}_n(K)) \cong K^\times$. Thus, automorphisms of this form are outer. This is, however, the only “new” source of automorphisms that we did not see when studying $\mathrm{GL}_2(\mathbb{F}_p)$. The below theorem makes this precise.

Theorem 4 (Structure of $\mathrm{Aut}(\mathrm{GL}_n(K))$). Let K be a field. Every automorphism of $\mathrm{GL}_n(K)$ can be expressed as the composition of an inner automorphism, a radial automorphism, and a field automorphism.

Proof. See, for example, Dieudonné’s *La géométrie des groupes classiques*, section 4.1 [Die71]. $\square$

References

- [Die71] Jean A. Dieudonné. *La géométrie des groupes classiques*. Vol. Band 5. Ergebnisse der Mathematik und ihrer Grenzgebiete [Results in Mathematics and Related Areas]. Troisième édition. Springer-Verlag, Berlin-New York, 1971, pp. viii+129 (cit. on p. 6).